Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

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1. Objective

To build a fully connected autoencoder, a convolutional autoencoder (CAE), a denoising CAE and a variational autoencoder (VAE) on MNIST; compare their reconstructions with MSE, MAE and SSIM; visualize the latent space; and generate new digits with the VAE.

2. Dataset and Experimental Setup

MNIST $28 \times 28 \times 1$ grayscale digits were scaled from $[0, 255]$ to $[0, 1]$. Labels were used only for plots; the clean image was always the target: $x \rightarrow$ Encoder $\rightarrow z \rightarrow$ Decoder $\rightarrow \hat{x}$.

Setting	Used	Manual
Training / validation / test images	10,000 / 2,000 / 2,000	10,000 / – / 2,000
Input (FC-AE / CAE)	$x \in \mathbb{R}^{784}$ / $x \in \mathbb{R}^{28\times28\times1}$	same
Batch tensor	$X \in \mathbb{R}^{128\times1\times28\times28}$	example: 32
Optimizer, learning rate	Adam, 10^{-3}	Adam, 10^{-3}
Batch size, epochs	128, 20	128, 20
AE loss	Binary cross-entropy	BCE
SSIM	Mean over test images, data range 1, 7×7 window	–
Framework, seed	PyTorch 2.10 (CPU), seed 42	–

Test images were never used for training or model selection.

3. Overall Experimental Pipeline



Model	Architecture	Parameters
FC-AE	784 – 128 – 32 – 16 – 32 – 128 – 784; ReLU hidden, sigmoid output	211,040
CAE	Conv32 – Pool – Conv64 – Pool – Conv64 ($64 \times 7 \times 7$ latent) – Up – Conv32 – Up – Conv1 (sigmoid)	74,497
Denoising CAE	Same as CAE; noisy input, clean target	74,497
VAE ($d_z = 2$)	Conv32/s2 – Conv64/s2 – Flatten – $\mu, \log \sigma^2 \in \mathbb{R}^2$ – Dense – ConvTranspose – $28 \times 28 \times 1$	139,461

4. Fully Connected Autoencoder

Plot 1: Original versus Reconstructed Images

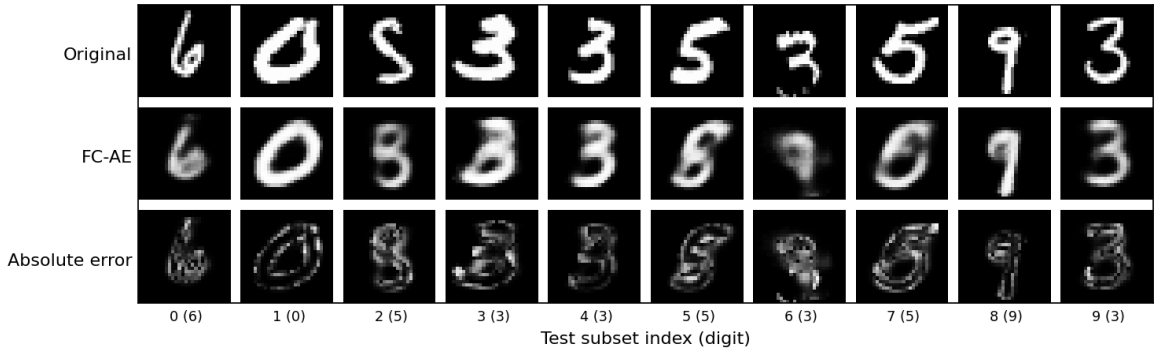


Figure 1: Ten test images, FC-AE ($d_z = 16$) reconstructions and absolute error.

Inference: Overall digit shape is kept for easy samples (0, 9, 3), but thin strokes are blurred and some 5s turn into 8-like shapes. The error row shows the loss is concentrated on stroke edges and unusual handwriting.

Plot 2: Training and Validation Reconstruction Loss

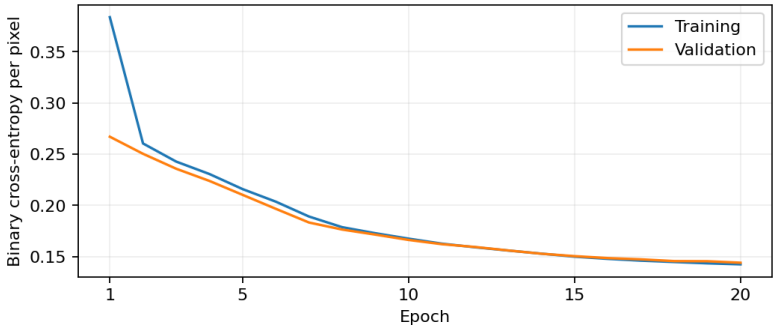


Figure 2: FC-AE binary cross-entropy per pixel.

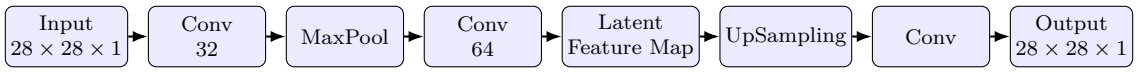
Inference: Training BCE fell from 0.3837 to 0.1424 and validation BCE from 0.2670 to 0.1440. The curves stay together, so the model converged with no sign of overfitting.

Reconstruction Metrics

$MSE = \frac{1}{N}\sum(x_i - \hat{x}_i)^2$, $MAE = \frac{1}{N}\sum|x_i - \hat{x}_i|$, SSIM compares local structure (1 = identical).

Metric (FC-AE, $d_z = 16$)	Value
Test MSE	0.027172
Test MAE	0.069386
Mean SSIM	0.692002

5. Convolutional Autoencoder



Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.027172	0.069386	0.692002	211,040
Convolutional AE	**0.002742**	**0.015532**	**0.974080**	74,497

Plot 3: Reconstruction Comparison

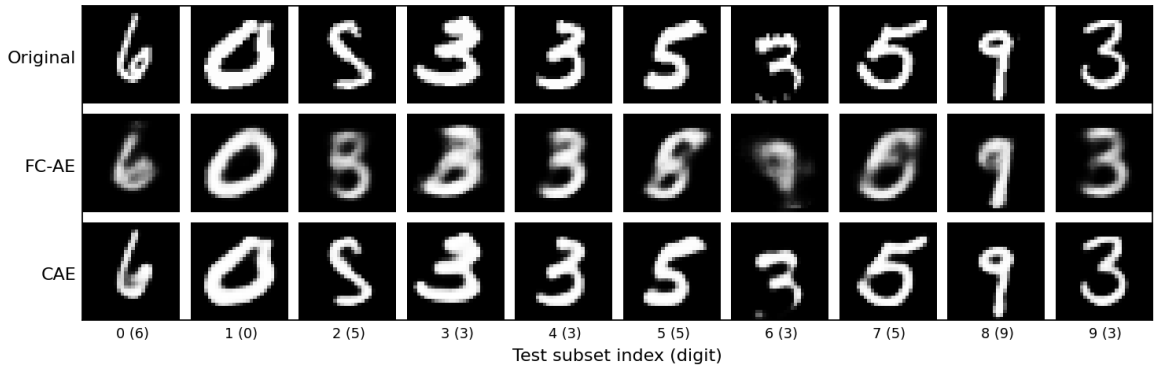
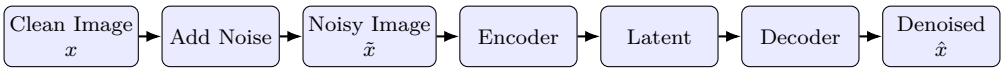


Figure 3: Original → FC-AE reconstruction → CAE reconstruction.

Inference: The CAE keeps sharp strokes and cut MSE by about 10× with one third of the parameters. Convolution keeps neighbouring pixels together and shares filters across positions, so local stroke structure survives. Its $64 \times 7 \times 7$ latent (3,136 values) is also much larger than the FC-AE's 16 values.

6. Denoising Autoencoder



Two corruptions were used: Gaussian $\tilde{x} = \text{clip}(x + n, 0, 1)$, $n \sim \mathcal{N}(0, \sigma^2)$, and salt-and-pepper, where a fraction p of pixels is set to 0 or 1. The CAE was trained on $\sigma = 0.2$ Gaussian or $p = 0.10$ salt-and-pepper inputs with clean targets.

Plot 4: Clean vs. Noisy vs. Denoised

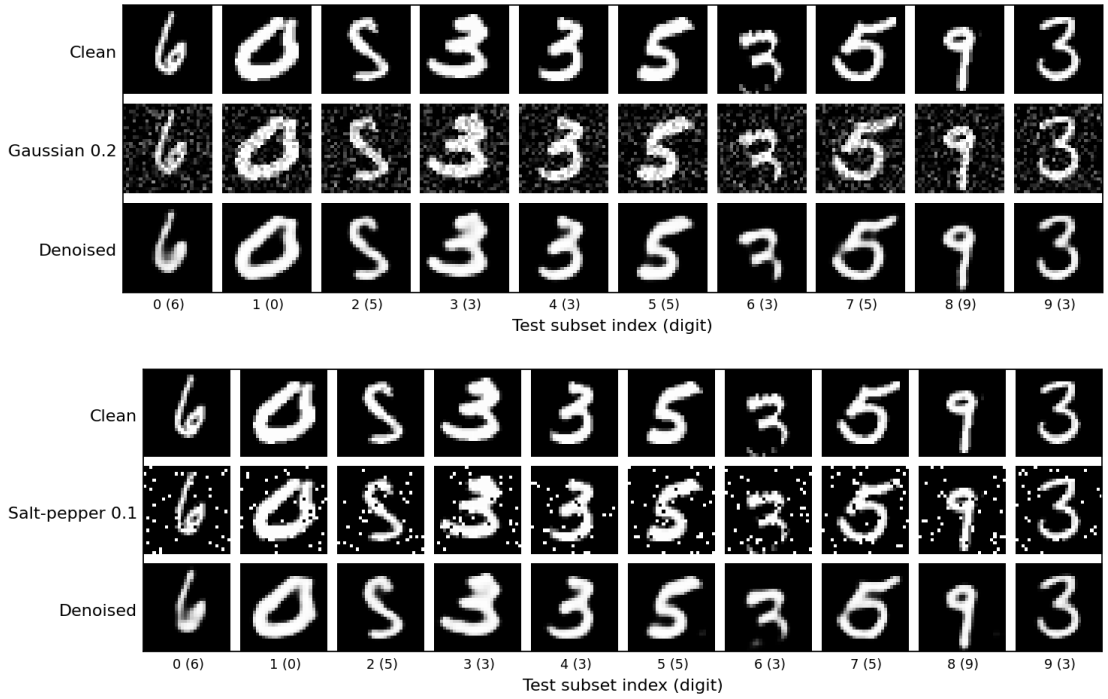


Figure 4: Gaussian $\sigma = 0.2$ (top) and salt-and-pepper $p = 0.10$ (bottom).

	Noisy input			Denoised output		
Noise Level	MSE	MAE	SSIM	MSE	MAE	SSIM
Gaussian $\sigma = 0.1$	0.005422	0.044053	0.700865	0.003518	0.017971	0.961530
Gaussian $\sigma = 0.2$	0.021167	0.086510	0.616586	0.004493	0.021158	0.948878
Gaussian $\sigma = 0.3$	0.046530	0.127839	0.531867	0.007470	0.031121	0.863364
S&P $p = 0.05$	0.024017	0.024944	0.725749	0.004240	0.019763	0.957220
S&P $p = 0.10$	0.048041	0.049896	0.598735	0.005150	0.022092	0.945470
S&P $p = 0.20$	0.096425	0.100154	0.466041	0.008290	0.029394	0.899596

Plot 5: Noise Level vs. Reconstruction Quality

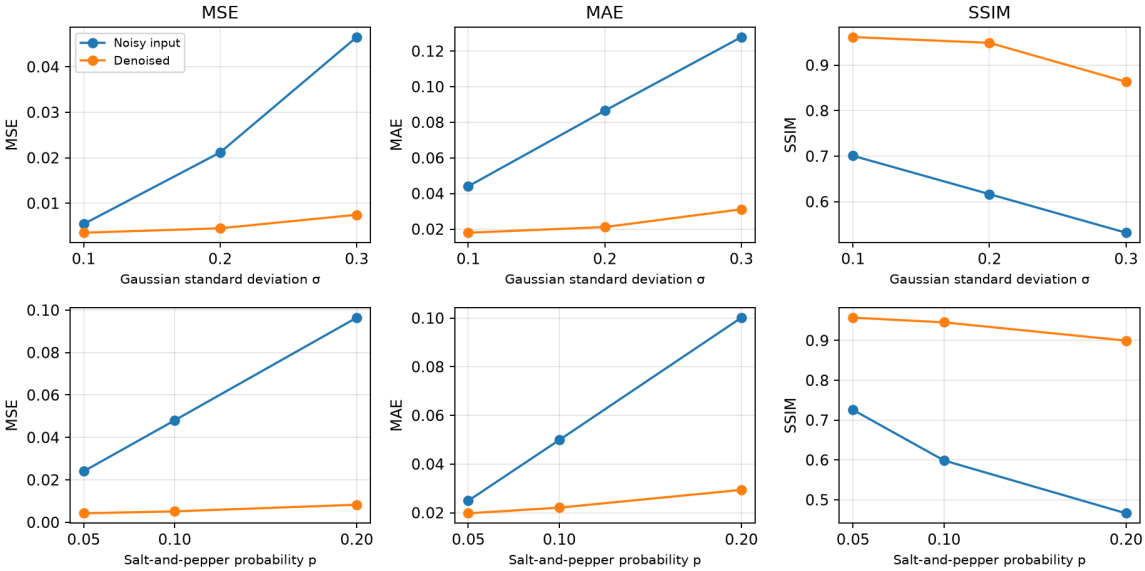
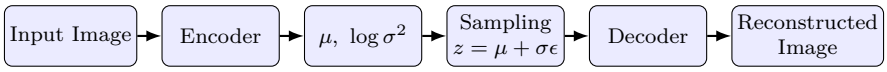


Figure 5: MSE, MAE and SSIM against noise level for Gaussian (top) and salt-and-pepper (bottom) noise.

Inference: As σ rose from 0.1 to 0.3, denoised SSIM fell from 0.962 to 0.863 because less stroke information survives. The denoised output was far better than the noisy input at every level, and the main digit structure was recovered for both noise types.

7. Variational Autoencoder



The encoder outputs $q_\phi(z|x) = \mathcal{N}(\mu(x), \mathrm{diag}(\sigma^2(x)))$ with $z \in \mathbb{R}^2$. A sample is drawn with the reparameterization trick $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$. The loss is

$$\mathcal{L}_{VAE} = \underbrace{\mathrm{BCE}(x, \hat{x})}_{\text{summed over pixels}} + \beta\, D_{KL}, \qquad D_{KL} = -\tfrac{1}{2}\sum_j \left(1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2\right),$$

with prior $p(z) = \mathcal{N}(0, I)$ and $\beta = 1$.

Plot 6: VAE Latent Space

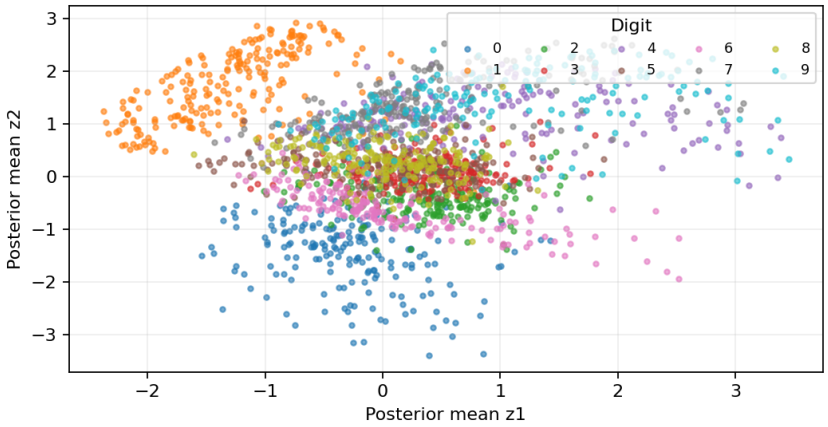


Figure 6: Posterior means (z_1, z_2) of the 2,000 test images, coloured by digit label.

Inference: Digits 0 and 1 form clear separate regions, while 4, 7, 9 and 3, 5, 8 overlap in the centre. The latent means have average (0.12, 0.38) and spread (0.94, 1.08), close to the $\mathcal{N}(0, I)$ prior, which the KL term enforces.

Plot 7: Randomly Generated Images

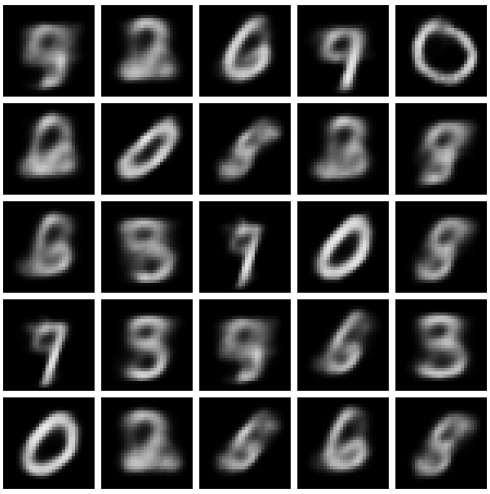


Figure 7: 25 images decoded from $z \sim \mathcal{N}(0, I)$.

Inference: Most samples look like handwritten digits (0, 2, 6, 7, 9 are recognizable) and there is clear variety, but several are blurred or ambiguous between 3, 5 and 8, matching the overlap in Plot 6.

Plot 8: Latent Space Interpolation

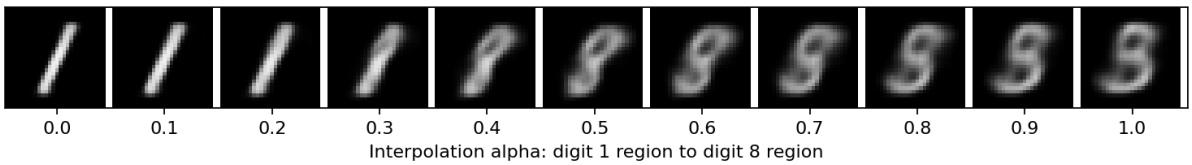


Figure 8: Decoded $z(\alpha) = (1-\alpha)z_A + \alpha z_B$ for $\alpha = 0, 0.1, \ldots, 1$, from a digit-1 code to a digit-8 code.

Inference: The image changes gradually from 1 to 8 (mean MSE between neighbouring images 0.0020), so the latent space is smooth; a few middle images are unclear blends.

VAE Training and Validation Loss

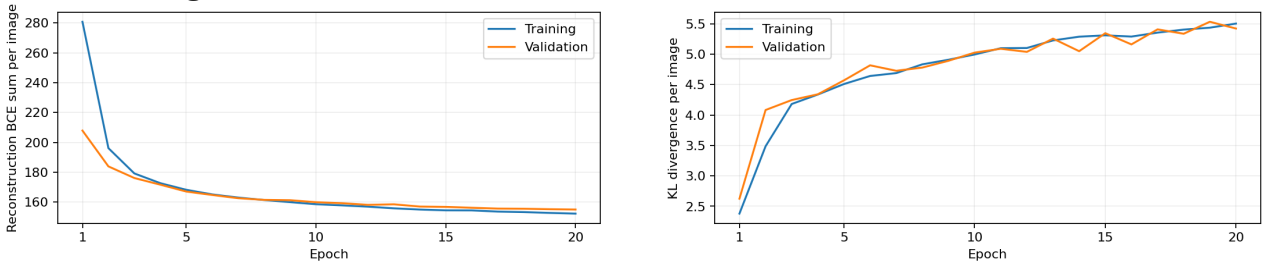


Figure 9: Reconstruction BCE (left) and KL divergence (right) per image.

Inference: Validation reconstruction loss fell from 207.9 to 155.0 while KL rose from 2.6 to 5.4: the encoder spends more of the latent capacity as it learns to reconstruct. Training and validation curves stay close, so there is no overfitting.

VAE Metric ($d_z = 2$, test set)	Value
Reconstruction Loss (BCE / image)	155.0826
KL Loss	5.4195
Total Loss	160.5020
Test MSE	0.045632
Test MAE	0.106545
Mean SSIM	0.492553

Inference: A conventional autoencoder maps each image to one fixed code, so gaps in its latent space decode to nothing meaningful. The VAE maps each image to a distribution near $\mathcal{N}(0, I)$, which costs reconstruction sharpness (SSIM 0.49) but allows sampling and smooth interpolation.

8. Reconstruction Comparison Across Models

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.027172	0.069386	0.692002	211,040
Convolutional AE	0.002742	0.015532	0.974080	74,497
Denoising CAE (noisy $\sigma = 0.2$ input)	0.004493	0.021158	0.948878	74,497
Denoising CAE (clean input)	0.003273	0.017009	0.965008	74,497

9. Reconstruction Error Distribution

Per-image error $e_i = \frac{1}{D}\sum_j (x_{ij} - \hat{x}_{ij})^2$ was computed for the FC-AE on all 2,000 test images.

Plot 9: Distribution of Reconstruction Errors

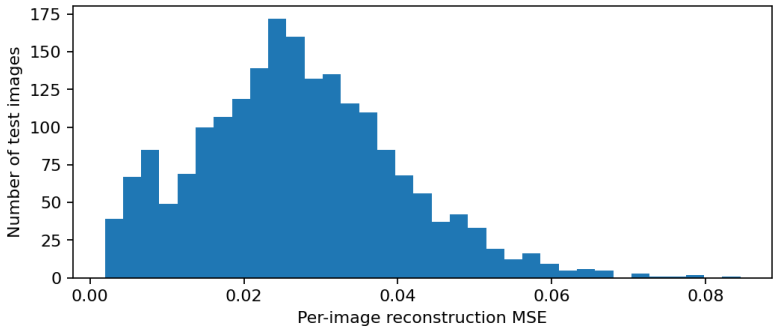


Figure 10: Histogram of per-image FC-AE reconstruction MSE.

Inference: Errors are concentrated between 0.018 and 0.035 (median 0.026), with a long right tail up to 0.085 formed by a few hard images.

High-Error Image Analysis

Sample	Original	Reconstruction	Test index	Digit	MSE	SSIM
1			5888	4	0.084535	0.427788
2			3778	5	0.078507	0.210007
3			1601	3	0.077531	0.314963
4			8196	6	0.075340	0.399861
5			3841	4	0.073193	0.325368

Inference: These digits have thick, slanted or unusual strokes, so the 16-value code snaps them to a more common digit ($4 \to 6$, $5 \to 9$, $6 \to 0$). High MSE matched visibly wrong reconstructions and low SSIM, so the error measures agree with visual quality.

10. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder ($d_z = 16$)	0.027172	0.069386	0.692002	211,040	4.51
Conv. Autoencoder	0.002742	0.015532	0.974080	74,497	76.86
Denoising CAE ($\sigma = 0.2$ input)	0.004493	0.021158	0.948878	74,497	75.24
VAE ($d_z = 2$)	0.045632	0.106545	0.492553	139,461	104.95

Time is total training time for 20 epochs on CPU, including validation.

11. Additional Latent-Dimension Study

Latent Dimension	MSE	MAE	SSIM	Parameters
2	0.067606	0.150665	0.177235	210,130
8	0.028763	0.072645	0.681657	210,520
16	0.027172	0.069386	0.692002	211,040
32	0.025608	0.066786	0.706838	212,080

Plot 10: Latent Dimension vs. Reconstruction MSE

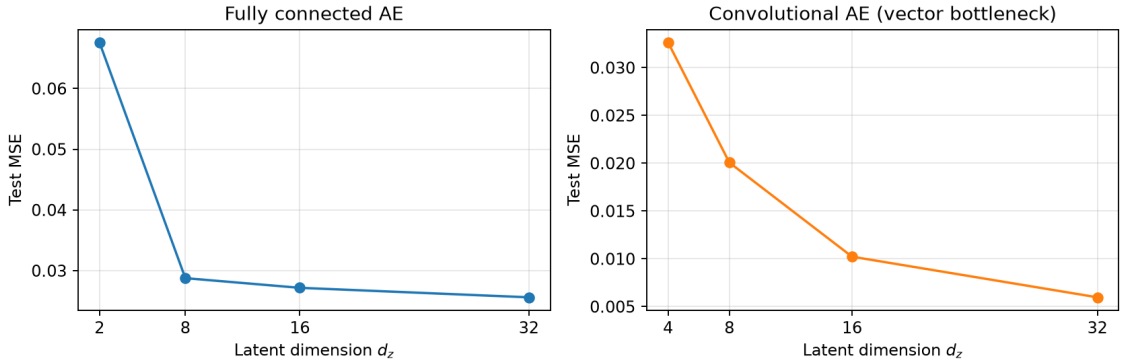


Figure 11: Test MSE against latent size for the FC-AE (left) and a CAE with a vector bottleneck (right).

Inference: A larger latent keeps more detail and lowers MSE, at the cost of less compression and more parameters. Gains flatten for the FC-AE beyond $d_z = 8$. At $d_z = 2$ the FC-AE's ReLU code collapsed to zero for every image, so it output one blurred average digit.

12. Additional Exercises

1. CAE with latent dimensions 4, 8, 16 and 32

Latent Dimension	MSE	MAE	SSIM	Parameters	Time (s)
4	0.032648	0.080543	0.658076	102,725	85.70
8	0.020067	0.053428	0.789090	127,817	87.09
16	0.010224	0.033164	0.894051	178,001	87.74
32	0.005971	0.023702	0.938890	278,369	91.40

Inference: MSE fell steadily as the vector bottleneck grew (right panel of Plot 10).

2–3. Noise type and training noise level

Model (training noise)	Test σ	MSE	MAE	SSIM
DAE, $\sigma = 0.2$	0.1	0.003518	0.017971	0.961530
DAE, $\sigma = 0.2$	0.2	0.004493	0.021158	0.948878
DAE, $\sigma = 0.2$	0.3	0.007470	0.031121	0.863364
DAE, $\sigma = 0.3$	0.1	0.004250	0.019739	0.952989
DAE, $\sigma = 0.3$	0.2	0.004970	0.021821	0.943955
DAE, $\sigma = 0.3$	0.3	0.006552	0.026186	0.923018

Inference: Training on stronger noise raised SSIM at $\sigma = 0.3$ (0.863 $\rightarrow$ 0.923) but slightly lowered

it at $\sigma = 0.1$. Under the same architecture, salt-and-pepper and Gaussian noise were both removed well (Section 6).

4. UpSampling vs. transposed-convolution decoder

Decoder	MSE	MAE	SSIM	Parameters	Time (s)
UpSampling2D	0.002742	0.015532	0.974080	74,497	76.86
Conv2DTranspose	0.002303	0.014238	0.977181	89,057	58.95

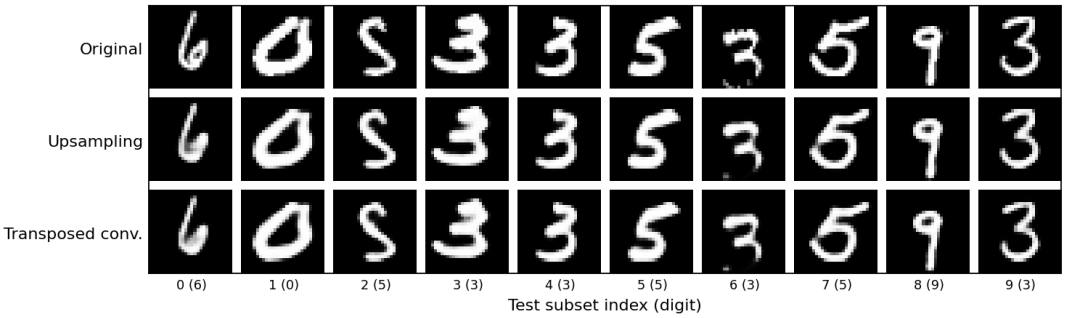


Figure 12: Original, UpSampling CAE and transposed-convolution CAE reconstructions.

Inference: Transposed convolutions learn their own upsampling, giving slightly lower error for 14,560 extra parameters; the images are visually almost identical.

5–6. VAE latent dimension and KL weight

Model	Recon. BCE	KL	Total	MSE	MAE	SSIM
VAE $d_z = 2$, $\beta = 1$	155.0826	5.4195	160.5020	0.045632	0.106545	0.492553
VAE $d_z = 8$, $\beta = 1$	99.2028	15.9193	115.1221	0.019473	0.053841	0.786753
VAE $d_z = 2$, $\beta = 4$	160.8051	3.4339	174.5406	0.046787	0.111755	0.464868

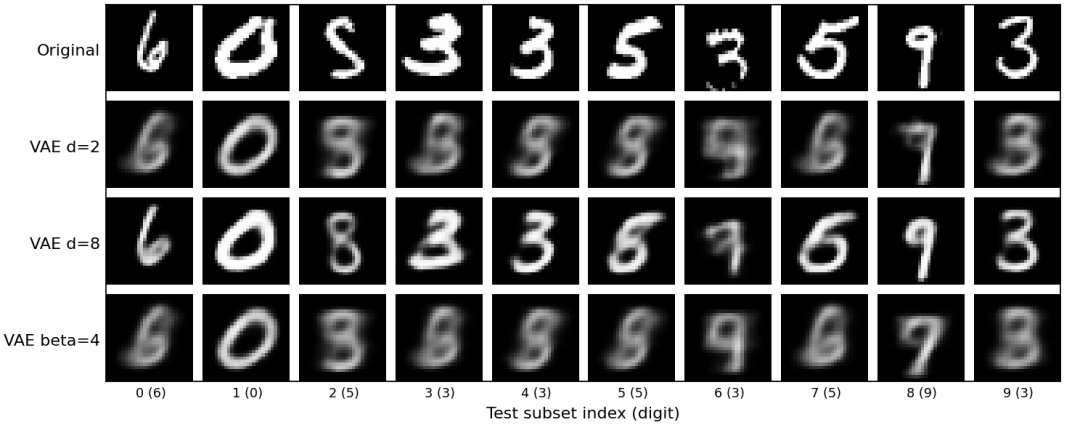


Figure 13: Reconstructions from the three VAE variants.

Inference: $d_z = 8$ reconstructs far better (SSIM 0.79 vs. 0.49). Raising β to 4 pulls the posterior closer to the prior (lower KL) and blurs the images further.

7–8. Larger sample grid and 8-D latent space

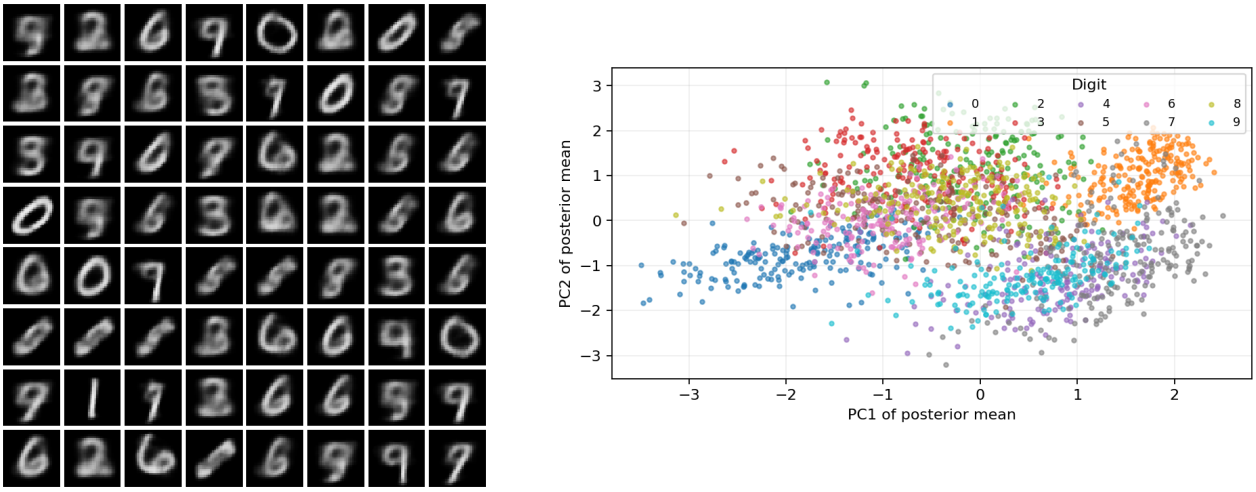


Figure 14: 64 generated samples from the 2-D VAE (left) and a 2-component PCA view of the 8-D VAE latent means (right).

Inference: The larger grid shows good variety but also repeated and ambiguous shapes. The PCA view keeps only 36.7% of the 8-D variance, yet digit groups (0, 1, 7) are still visible. Interpolation between two digit regions is shown in Plot 8.

13. Discussion Questions

No.	Question	Answer
1	What is an autoencoder?	A network trained to reconstruct its own input through a bottleneck.
2	Purpose of the encoder?	Compresses the image into a latent representation.
3	Purpose of the decoder?	Rebuilds the image from that representation.
4	What is a latent representation?	The compact code the encoder learns.
5	Why is the latent smaller than the input?	It forces compression instead of simple copying.
6	What if the latent is too small?	Detail is lost and error rises ($d_z = 2$ FC-AE).
7	What if the latent is increased?	More detail is kept, but compression drops and gains flatten.
8	Why does a CAE preserve spatial information?	Filters work on neighbouring pixels and share weights across positions.
9	Autoencoder vs. denoising autoencoder?	A DAE gets corrupted input but must output the clean image.
10	Why is the clean image the target?	It teaches the model to remove noise, not copy it.
11	What happens as noise increases?	Less information survives, so recovery gets worse.
12	Why use MSE, MAE and SSIM?	MSE punishes large errors, MAE gives average error, SSIM measures structure.
13	What is a VAE?	An autoencoder whose encoder outputs a probability distribution.
14	How does a VAE differ from an AE?	It learns a distribution per input, not one fixed code.
15	Why learn a distribution?	It gives a smooth, sampleable latent space for generation.
16	What is the reparameterization trick?	Writing $z = \mu + \sigma\epsilon$ so gradients flow through μ and σ.
17	Why sample ϵ from $\mathcal{N}(0, I)$?	Scaling and shifting it gives any required Gaussian sample.
18	Role of KL divergence?	Keeps each posterior close to the prior.
19	Which prior is used?	Standard normal, $p(z) = \mathcal{N}(0, I)$.
20	Why use a 2-D latent space?	It can be plotted directly.
21	What does interpolation show?	Whether the latent space changes smoothly.
22	Why can a VAE generate new images?	Its decoder turns any prior sample into an image.
23	Why are some samples unclear?	Overlapping digit regions and the KL penalty blur outputs.
24	Reconstruction vs. generation?	Reconstruction encodes a real image; generation starts from a random z.
25	Why keep test images unseen?	To measure generalization fairly.

14. Conclusion

The CAE reconstructed MNIST best (SSIM 0.974) with the fewest parameters, far ahead of the 16-value FC-AE (0.692). The denoising CAE recovered digit structure from both Gaussian and salt-and-pepper noise, with quality falling as noise grew. The 2-D VAE traded reconstruction sharpness for a smooth, prior-shaped latent space that supports sampling and interpolation; a larger latent or a smaller β improves its reconstructions.

15. Source Code

GitHub repository	`https://github.com/Joshua-Divide/LAB`
Notebook	`Lab 7/Experiment 7.ipynb`